Impacts of Public Early Childhood Education in Chile on Maternal Fertility and Labor Market Outcomes

Short-run effects — progress update

Andrés Hojman Marigen Narea Jorge Rodríguez Pedro Carneiro Micaella Magnasco

Pontificia Universidad Católica de Chile

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Where we are

What we do

Question: does opening a public preschool nearby change mothers' labor supply and fertility?

Variation: 447 neighborhood units (UVs) open their *first* JUNJI/Integra preschool in 2015–2024 vs. 4,145 that never do

Design: staggered event study at the UV level, Sun–Abraham estimator (Baker et al., *JEL* 2026)

Data: woman-level panel linking administrative registries — RSH/FPS, SII/AFP earnings, Civil Registry births, preschool enrollment

What we find so far

First stage: enrollment **+2pp** (33% → 35%), from the opening year, no pre-trends; strongest at ages 2–3

Formal employment: **+1pp** from year 4 on; pre-trends not significant but rising

Earnings (mothers who work): **suggestive +2–4%**, not yet significant

Months worked: no effect so far

Fertility: no effect so far — a fairly precise zero

Cleaner control group: same results

Project status

Done: data, treatment, balance, first stage, first labor and fertility estimates

In progress: 2015 cohort check, earnings including non-working mothers

Next: pre-trend robustness, mechanisms, writing

Research question

Question

What are the effects of the public expansion of early childhood education in Chile on mothers' labor market participation and fertility?

What we know

Mixed evidence: positive effects on maternal employment in Quebec and Argentina, close to zero in Norway

Recent Latin American evidence comes from **lotteries and experiments** (Rio de Janeiro, Nicaragua)

Less is known about a **national, decade-long public expansion** observed in administrative data

Our contribution

First study to use neighborhood-unit (UV) variation: preschools geolocated to the UV and linked to individual administrative records

Fertility as a second margin, alongside labor supply

What is a UV? *Unidad Vecinal*: the official sub-municipal territorial unit — ∼6,900 UVs vs. 346 municipalities.

⇒ **Finer geography: a sharper measure of exposure and more variation to identify the effect.**

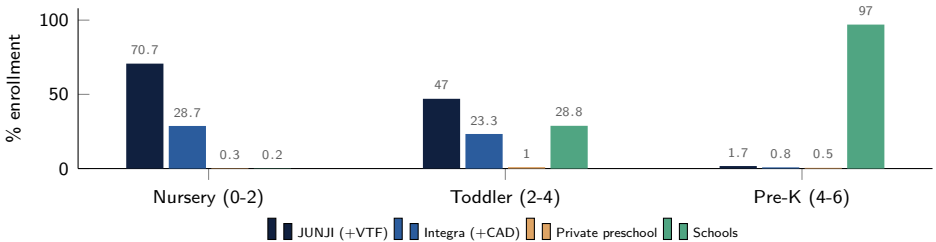
Quebec: Baker et al. (2008); Argentina: Berlinski & Galiani (2007); Norway: Havnes & Mogstad (2011); Rio: Attanasio et al. (2025); Nicaragua: Hojman & López Bóo (2022).

Public provision expanded sharply between 2015 and 2024

Mixed provision: **public** (JUNJI, Integra, municipal/SLEP) vs. **private** (subsidized/paid)

700+ new public preschools opened between 2015 and 2024

Net attendance rate (ages 0–5) rose from **50% to 55%** over the period, despite pandemic-era setbacks



Source: authors' elaboration based on Figura 5, Matrícula Oficial 2023, Centro de Estudios MINEDUC.

⇒ **Ages 0–4: JUNJI and Integra are the main providers — our focus.**

Identification: staggered-adoption event study

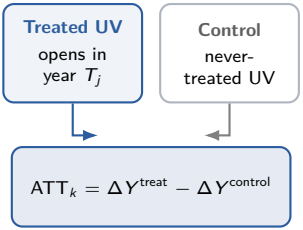
Design: staggered-adoption difference-in-differences with event study

Unit of treatment: the UV, the year it opens its first public preschool

Unit of analysis: **UV** $\times$ **year** panel, built from a woman $\times$ year panel linked to each woman's UV of residence

Fixed effects: **UV + year** (baseline); woman FE on the woman $\times$ year panel as a **robustness check**

Estimand: ATT of UV-level treatment (Baker et al., *JEL* 2026) — an **intent-to-treat effect of access**, not individual take-up



$\Rightarrow$ **We compare changes in UVs that get their first preschool vs. UVs that never do — an intent-to-treat effect of access.**

A UV is treated when it opens its first preschool

Treatment definition

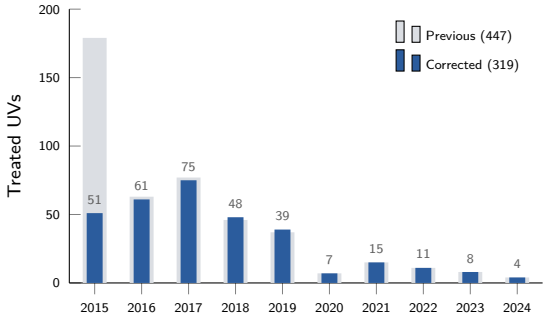
A UV is **treated** the year it opens its first public preschool (JUNJI/Integra) — goes from 0 to 1+ centers.

Group	Definition	N (UVs)	%
Excluded	Already had 1+ center in 2014 (out of design)	2,394	34.8%
Never-treated	0 in 2014, stays at 0 through 2024	4,174	60.6%
Treated	0 in 2014, opens first center 2015–2024	319	4.6%
Total universe		**6,887**	100%

Source: `base_cobertura_cp.dta`, corrected opening years (Oct 21). Design sample (`muestra_t1==1`) = 4,493 UVs. Previous version: 447 treated, 2,265 excluded. Counts in the RIS samples (UVs with women) pending re-run.

$\Rightarrow$ **We identify the effect of the first preschool (0 $\rightarrow$ 1): 319 treated vs. 4,174 never-treated.**

Treatment cohorts by year



2015 cohort corrected: 179 → **51 UVs** — in 127 UVs the "opening" was a center already operating in 2014 (next slide)

Other cohorts barely change; **2017 is now the largest** (75 UVs)

59% open in 2015–2017; only 45 UVs open after 2019

The 2015 cohort still has **only one pre-period** (2014)

Source: `base_cobertura_cp.dta`, corrected opening years (Oct 21). 319 treated UVs (0 centers in 2014).

⇒ **Openings are spread over 2015–19, no longer concentrated in 2015.**

Resolved: most 2015 "openings" were centers already operating

Problem: JUNJI opening dates come from records that start in March 2015 → centers with no earlier record were dated **01-03-2015** (84% of 2015 openings)

Evidence: older transparency requests give **center × year panels** (JUNJI 1996–2023, Integra 2001–2024); JUNJI centers are flat at ∼3,150 a year in 2011–2016

Fix: opening year = first year the center appears in the panels

JUNJI centers "opening" in 2015

Recorded date	Centers	Already there
01-03-2015	444	**427 (96%)**
Other 2015 date	108	15 (14%)

UVs	Before	After
Treated	447	**319**
2015 cohort	179	**51**
Excluded (center in 2014)	2,265	2,394
Never-treated	4,175	4,174

Pending: re-run everything with the corrected treatment

All results that follow still use the previous treatment (447 treated UVs). The "without 2015 cohort" estimates are the closest preview.

⇒ **The 2015 cohort was mostly a data artifact: 127 of its 179 UVs already had a center in 2014.**

Empirical specification and estimator

$$Y_{ut} = \lambda_u + \lambda_t + \gamma\, \overline{\text{Age}}_{ut} + \sum_{k \neq -1} \beta_k\, \mathbf{1}\{t - G_u = k\} + \varepsilon_{ut}$$

Y_{ut}: average outcome of women in UV u, year t (built from the woman $\times$ year panel). λ_u, λ_t: UV and year fixed effects; SE clustered by UV.

$\overline{\text{Age}}_{ut}$: average age of mothers in the UV-year (labor market specification).

β_k: effect k years from opening, relative to $k = -1$ (omitted); $k \leq -5$ binned.

Estimation strategy

Baseline TWFE as reference — but with staggered treatment, TWFE can use already-treated units as controls (Goodman-Bacon problem). Robust estimator: **Sun–Abraham** (`eventstudyinteract`), never-treated UVs as control group. Framework: Baker, Callaway, Cunningham, Goodman-Bacon & Sant'Anna (*JEL*, 2026).

Main threat: non-random placement of openings (JUNJI/Integra prioritizing UVs by care demand or socioeconomic profile). **UV FE** absorb permanent differences across UVs; **pre-period coefficients** ($k < -1$) test whether trends were already diverging.

$\Rightarrow$ **Main estimator: Sun–Abraham with never-treated UVs as controls; TWFE as reference.**

Data: what we have and how each sample is built

Sample	Source data	Key variables
Coverage (UV × year)	JUNJI/Integra transparency requests + UV shapefile (MDSF)	Number of centers, treatment status, opening cohort
Women panel (base)	RSH (2016–2024) + FPS (2014–2015)	UV of residence, UV population, child density
Labor market	+ Rentas (SII/AFC) + Cotizaciones (AFP)	Annual income, months worked, employment
Fertility	+ Civil Registry birth records (via RIS)	Birth in the year, children under 5
First stage (enrollment)	+ Matrícula parvularia (child-level)	Enrollment, enrollment rate

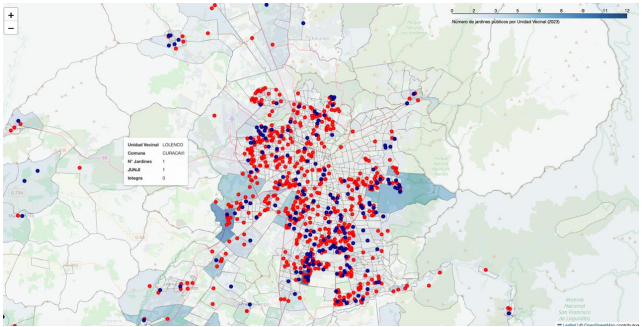
⇒ **All administrative registries, linked at the individual level — no survey self-reports.**

Every preschool is matched to its 2024 neighborhood unit

Each JUNJI/Integra preschool is **geolocated** and matched to the UV where it sits (spatial join)

UV boundaries **changed a lot** over time → we traced them and mapped everything to **2024 boundaries**

Preschools and women's UV of residence share **one common geography** across all years

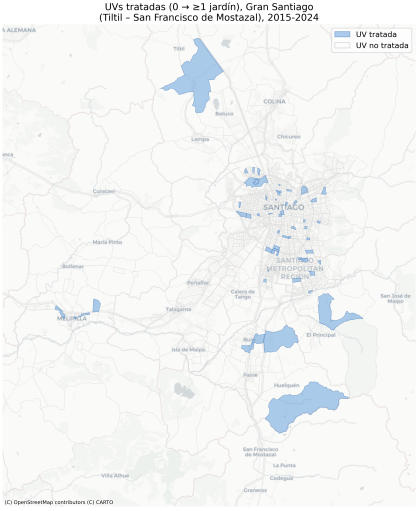


Región Metropolitana. ● JUNJI ● Integra Shading: public preschools per UV. Source: `base_cobertura_cp.dta`.

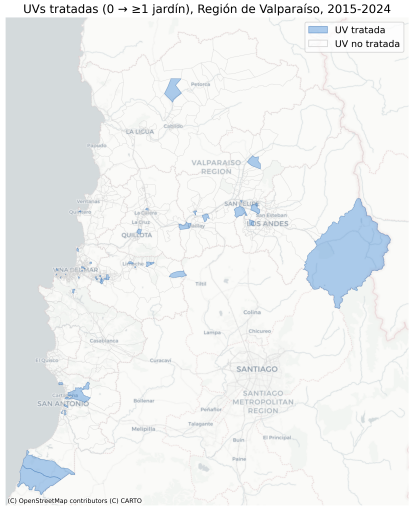
⇒ **One common geography: every preschool and every woman mapped to 2024 UV boundaries.**

▸ Interactive map 2014–2024

Región Metropolitana



Valparaíso



Illustrative examples. Shaded: treated UVs (0 → 1+ preschools), 2015–2024.

Who is in each sample, and what we measure

1. Samples (selected at the individual level)

Fertility: women of fertile age (18–49)

Labor market: women with a child <5 *that specific year* — not "ever had a child"

First stage: children of treatment-sample mothers, birth to birth+4

2. Outcomes (averaged at UV × year)

Fertility: share of women who had a child that year (Civil Registry birth records)

Labor market: average annual income, months worked, share employed, dependent vs. independent

First stage: enrollment rate

⇒ **Each outcome is measured on its relevant population, from administrative records, averaged at the UV × year level.**

Treated and never-treated UVs look alike — except in size

Labor market sample (mothers with a child <5) — covariate levels in 2014, before any opening

	Unweighted			Weighted by UV population			
Variable	Never-tr.	Treated	Norm. diff.	Never-tr.	Treated	Norm. diff.	Sig.[a]
Formal employment (share)	0.47	0.48	0.075	0.52	0.51	-0.021	–
Annual income (M CLP)	3.60	3.55	-0.033	3.95	3.89	-0.039	*
Months worked	8.14	8.10	-0.024	8.44	8.37	-0.058	*
Age	29.24	29.25	0.008	29.26	29.29	0.026	
N. children <5	1.11	1.11	0.045	1.11	1.11	0.049	
Hours worked	37.07	37.68	0.099	36.55	36.81	0.068	
Works (share)	0.45	0.45	0.023	0.48	0.48	-0.006	
Education (code)	4.98	5.02	0.028	5.20	5.18	-0.020	
UV population	1126.79	2011.08	**0.410**	3530.92	5287.26	**0.340**	***
Child density 0–5 UV	0.06	0.06	0.155	0.06	0.06	0.139	***
N (UVs)	3,909 / 422			3,909 / 422			

[a] t-test at the woman level (* p<0.05, *** p<0.01); for reference only; normalized differences prioritized (concern threshold: 0.25); not computed for formal employment (–).

The UV population gap is **expected**: the State prioritizes new preschools where need and population are larger. Same pattern in the fertility sample (App. A4).

⇒ **Selection is in *where* preschools open (UV size), not in *who* lives there → UV fixed effects.**

About 1 in 3 mothers moves across UVs — equally in treated and control UVs

How many move?

33% of mothers change UV at least once while in the sample

7–12% per year

2022: 27% — the year codes were standardized: likely an artifact, not real moves

Do they move toward new preschools?

Group	Moves (% of mother-years)
Treated	12.6
Never-treated	12.3
Excluded	11.9

How we assign a UV to each woman: RSH UV codes are not consistent across years; they are standardized only from 2022. For 2017–2021 we decoded them backwards by following the same women to the next year (majority rule) to place everyone on 2024 boundaries. → **Part of the measured moves may be decoding errors, not real moves** (e.g., the 2022 jump).

Real moves dilute UV-level effects → we also follow each mother from her first UV (woman fixed effects, below).

⇒ **Mothers move often, but not toward UVs that open a preschool.**

Moves measured with 2024 UV boundaries, the same map that defines treatment. Year-to-year changes from 2017 on (2014–2015 UVs are proxied by 2016). 33% = among mothers observed in at least two comparable years.

Results: baseline specification

Results in the next slides use the previous definitions

All women in the RSH — no vulnerability restriction

Labor market: mothers with a child <5 *that specific year*

First stage: children of those mothers; **fertility**: women aged 18–49

UV-level estimates, all treatment cohorts (2015 included)

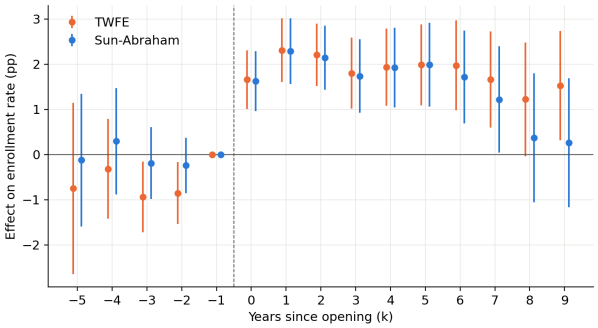
Suggestions from the last meeting

60% most vulnerable, mothers followed over time, individual-level estimates

Incorporated from slide 26 on

⇒ **First the baseline results, then what changes with the new samples.**

First stage: enrollment rises the year a preschool opens and stays higher



Y-axis: change in the enrollment rate (% of children 0–4) in treated vs. never-treated UVs, relative to the year before opening

Treated UVs: **33%** → **≈35%** (+2pp), from the opening year on

A treated UV has **≈135 children** aged 0–4 → **≈3 more enrolled**

No pre-trends

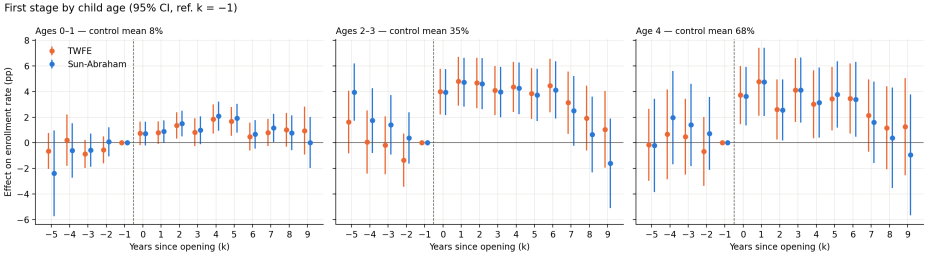
$k = 7$–9: only early cohorts (mostly 2015) → composition, not a fading effect

▸ Average effects (A6)

⇒ **A new preschool raises enrollment from 33% to ≈35%: ≈3 more children per UV.**

Weighted by number of children. Sun–Abraham and TWFE, 95% CI clustered by UV. Children aged 0–4.

First stage by child age: strongest at ages 2–3



Child age	Control mean	SA $k = 0$–6	Relative
Ages 0–1	8%	+1.3pp	+15%
Ages 2–3	35%	**+4.2pp**	+12%
Age 4	68%	+3.6pp	+5%

Ages 2–3: largest effect, immediate and stable

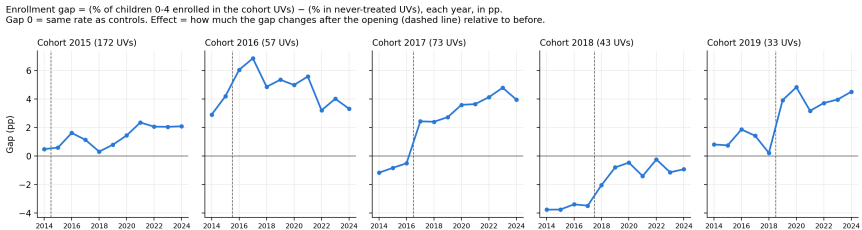
Ages 0–1: small in pp but largest in relative terms; grows over time

Age 4: also positive

Caution: ages are measured at different points in time across datasets (to check)

⇒ **Enrollment rises at all ages, most at ages 2–3.**

An open issue: the 2015 cohort



172 UVs — 41% of treated UVs, ∼half of the sample at $k = 6$, the only cohort at $k = 9$

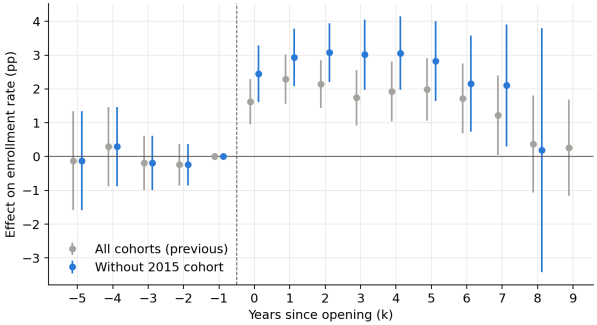
Its effect is **very small** (≈0.7pp in raw gaps) and it has only one pre-period (2014)

Hypothesis: registration error at the edge of the panel — preschools that already operated in 2014 but are recorded as 2015 openings.

Next: carefully review the coverage data provided by JUNJI/Integra.

⇒ **The 2015 cohort pulls the average effect down — possibly a data error to check.**

Without the 2015 cohort, the first stage is ∼50% larger



Same pre-periods (the 2015 cohort has none)

+2.7pp vs. +1.8pp: **≈4 more children** enrolled per UV instead of 3

Jumps at opening and stays stable

Earnings: the effect also roughly doubles without 2015

⇒ The 2015 cohort dilutes the first stage — but it remains small.

All mothers (first stage not yet estimated for the 60% most vulnerable). Average $k = 0$–7. Sun–Abraham, weighted by number of children, 95% CI clustered by UV.

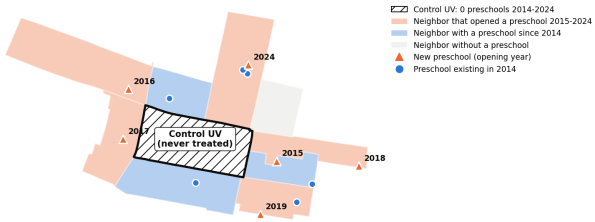
A cleaner control group: never-treated UVs can be exposed too

Concern: children from a never-treated UV can enroll in a new preschool next door → control enrollment rises → **effect underestimated**

We built a **neighbor map**: for every UV, which adjacent UVs had or opened a preschool (shared border, 2024 boundaries)

Example: this control UV has 9 neighbors — **5 open a preschool** during the period

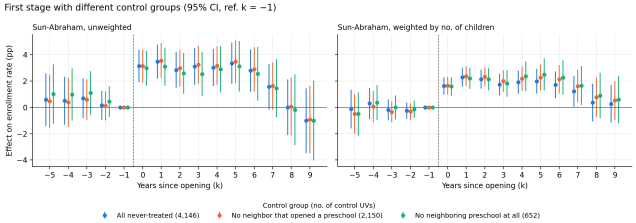
A 'never-treated' UV in La Cisterna
5 of its 9 neighboring UVs open a preschool in the period



⇒ **Goal: a larger first stage, comparing treated UVs with controls that could *not* use a neighbor's new preschool.**

* Neighbors only capture proximity to home. If mothers use preschools near their workplace, spillovers may reach UVs that are not adjacent.

A cleaner control group gives the same first stage



Control group	UVs	Median area	Control mean	SA weighted, $k = 0$–6
All never-treated (main)	4,146	2.8 km^2	33.2%	1.92
No neighbor opened a preschool 2015–24	**2,150**	3.8 km^2	32.7%	**2.14**
No neighboring preschool in any year	652	**6.7 km^2**	28.2%	2.13
Treated UVs	*422*	*1.5 km^2*		

Same effect with all three groups and clean pre-trends. The strictest group is not a good comparison: its UVs are $\sim$**4**$\times$ **larger** than treated ones (isolated areas) $\rightarrow$ preferred clean control: **no neighbor opened**.

$\Rightarrow$ **Spillovers to controls do not explain the small first stage.**

Why is the first stage small?

What we checked

2015 cohort: very small effect, possibly a data error → without it, ≈3pp

Cleaner control group: same ≈2pp → spillovers to controls are not the reason

What our outcome measures

Whether children **living** in the treated UV are enrolled in **any** preschool

A new preschool can fill up without moving our outcome

It raises enrollment only if it takes local children who were **not enrolled before**

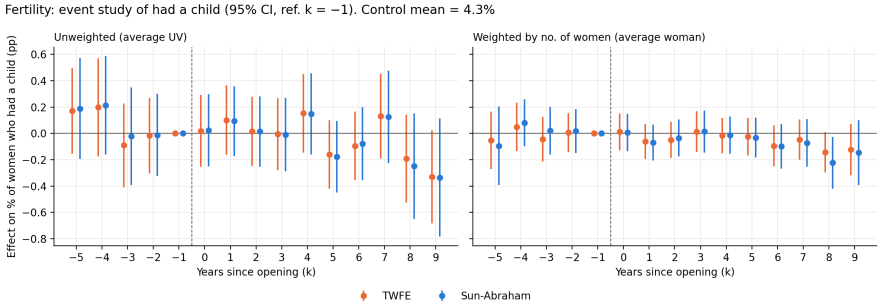
Substitution: local children already enrolled elsewhere switch to the new preschool → no net change

Mothers use other preschools, e.g. near their workplace in another UV → new places go to children from other UVs

Future work: enrollment records show which preschool each child attends → who fills the new preschool, where they live, and which preschools local mothers use.

⇒ **Enrollment of local children can move little, even if the new preschool fills up.**

Fertility: no effect yet



Treated and control UVs decline together; **no jump at opening**

Clean pre-trends

A **fairly precise zero**: rules out large effects

Still checking: cumulative outcome (having had a child)

⇒ **Expected: having a child depends on many factors, and a nearby preschool might weigh little.**

Women aged 18–49. Outcome: share who had a child in the year (control mean 4.3%). Controls for average age.

Mothers' formal employment rises by 1pp from the fourth year



Nothing at opening; **significant from** $k = 4$; no pre-trends ($p = 0.82$)

Why gradual? Two possible reasons

Who is in the sample: if the child is already 3–4 at opening, the mother barely uses it; later, more mothers had it since birth

Returning to work takes time

How to tell them apart:

Keep mothers until 2024 → no one leaves when her child turns 5

Woman FE → each mother compared with herself

If still gradual → it is the time to return to work

⇒ A new preschool raises mothers' formal employment by ≈1pp from the fourth year.

Weighted by number of mothers. Mothers with a child <5 that year. Employed = any taxable income or contribution.

Changes to the samples

1. Only the 60% most vulnerable (labor and fertility samples)

The population **eligible** for JUNJI/Integra preschools → removes women who could not be treated

Vulnerable = RSH bracket ≤60 (FPS score ≤13,500 in 2014–2015, new samples only)

Fixed per woman: her classification in the **first year she appears**, kept for all years → does not respond to the opening

2. Mothers stay in the labor sample (in progress)

Before: a mother is in the sample only **while** she has a child <5

Now: she enters the first year she has a child <5 and **stays until 2024**, even after the child turns 5 → the sample no longer changes with births and children aging out

⇒ **We now focus on eligible mothers and follow the same mothers over time.**

Results below: change 1 with RSH only and the previous mother definition. FPS and change 2: new samples being built.

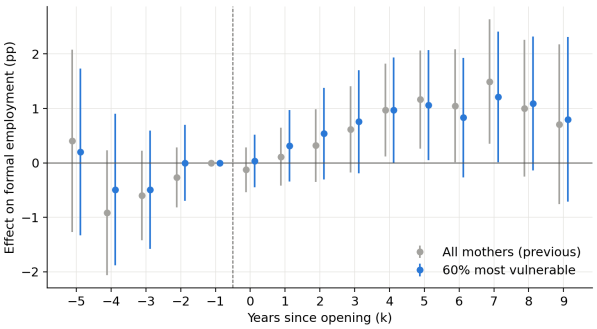
Where each change stands

	First stage	Labor market	Fertility
60% most vulnerable	pending	**done** (slides 28–30)	pending
Mothers followed until 2024	—	running	—
Woman FE, fixed UV	—	**preliminary** (slide 31)	—
Without the 2015 cohort	**done** (slide 20)	**done**: similar	pending
Earnings including zeros	—	**done** (slide 30)	—

Labor results for vulnerable mothers use RSH only and the previous mother definition. The first stage for their children comes next.

⇒ **Labor results already reflect the vulnerable sample; first stage and fertility come next.**

Employment: same effect among the 60% most vulnerable



No pre-trends (joint test $p = 0.79$)

Gradual, ≈+1pp from the 4th year; average **+0.8pp** on a 46% base (≈+1.6%), significant

Why so similar to all mothers? If the non-vulnerable third had no effect, we would expect ≈1.1pp (same effect, fewer mothers). But:

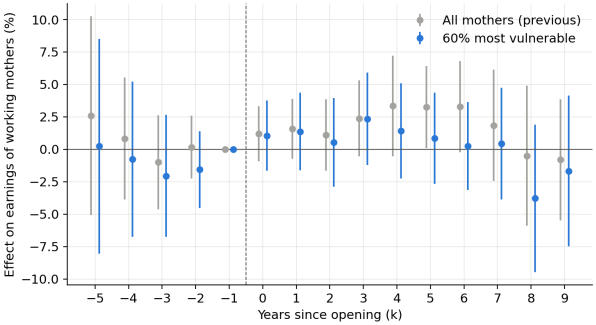
- Those excluded are RSH middle class, not rich; vulnerable mothers are 2/3 of the sample
- Eligibility is not strict: free places go to other families

Next: estimate the non-vulnerable separately.

⇒ **The employment effect holds in the eligible population.**

Sun–Abraham, weighted by number of mothers, 95% CI clustered by UV, ref. $k = -1$.

Working mothers do not earn more



In earnings: **only mothers who work**

No pre-trends; after opening, ≈**0** among vulnerable mothers (+0.3%, not significant)

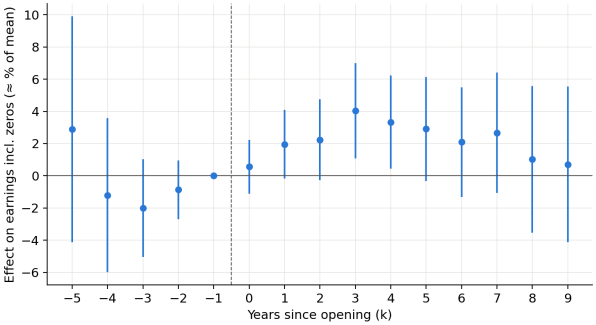
All mothers: slightly higher (≈+3% at $k = 3$–6), also not significant

Months worked: also no effect

⇒ **The earnings gain does not come from higher wages.**

Sun–Abraham, weighted by number of mothers with earnings, 95% CI clustered by UV, reference $k = -1$. Mothers with a child <5 that year.

Earnings rise because more mothers work, not because they earn more



Earnings **including zeros**: average over *all* mothers, those without earnings count as 0

Y-axis: 100 = average mother that year → **+2 points ≈ +2%**

Average after opening: ≈**+2%**, peak at $k = 3$–4 (significant at 10%)

∼**3/4** of it is explained by more mothers working

Among mothers who work: earnings and months worked **do not change**

⇒ **The preschool changes *whether* mothers work, not how much those already working earn.**

60% most vulnerable. Earnings relative to the year's mean (100). Sun–Abraham, weighted by number of mothers, 95% CI clustered by UV.

Woman fixed effects: a design to discuss

Approach

Compare **the same mother** before and after the opening → removes her fixed traits (e.g., attachment to work)

Treatment from her **UV before the opening**, kept if she moves (intention-to-treat)

Movers are **not dropped**: staying could itself respond to the preschool

Results so far (preliminary)

Employment: no pre-trends; gradual, significant from the 4th year, $\approx$+1.6pp by $k = 9$ — same pattern as UV level

Limitation

Does **not** address *where* and *when* preschools open — that happens at the UV level

Only helps if who lives in treated UVs changes after the opening

Open questions

Is woman FE worth it? Moves are similar in treated and control UVs (12.6% vs. 12.3%) → it may mainly serve as a robustness check

Fixed pre-opening UV or contemporaneous UV?

Should movers be kept or dropped?

⇒ **Same employment result following the same mothers — input welcome on the design.**

Women first observed in 2014–2015: UV proxied by 2016 (App. A2). To be re-estimated with the new mother sample.

Woman fixed effects: specification

$$E_{it} \;=\; \mu_i \;+\; \lambda_t \;+\; \sum_{k \neq -1} \beta_k \, \mathbf{1}\{\, t - G_{u_0(i)} = k \,\} \;+\; \varepsilon_{it}$$

```
eventstudyinteract empleo_formal d1-d4 d6-d15, cohort(g1_f) control_cohort(nunca_f)
    absorb(rut_inn anio) vce(cluster uv_fija)
```

E_{it}: formal employment of mother i in year t

μ_i: **woman** fixed effect; λ_t: year fixed effect

$u_0(i)$: **fixed UV** = first UV where the mother appears, before the opening

$G_{u_0(i)}$: year her fixed UV opens its first preschool $\rightarrow$ kept even if she moves

Sun–Abraham: cohorts by $G_{u_0(i)}$ (`g1_f`); controls = fixed UV never opens one (`nunca_f`)

$k \leq -5$ binned; ref. $k = -1$; SE clustered by fixed UV

Sample: vulnerable mothers whose fixed UV had no preschool in 2014, first observed before the opening

$\Rightarrow$ **Each mother is compared with herself, treated by the UV where she lived before the opening.**

Preliminary run: ≈5M mother-years, 1,634 clusters; estimated on an earlier version of the new mother sample, to be re-estimated.

Running and pending

Running

New samples: vulnerable mothers followed until 2024; fertility and first stage for the 60% most vulnerable

Woman fixed effects with the new sample: average effect and SE

Pending

Re-run everything with the corrected treatment (319 treated UVs): samples, balance, first stage, employment, earnings, fertility

First stage among the 60% most vulnerable

All outcomes with the new samples

Who fills the new preschool? Enrollment in JUNJI/Integra vs. other providers

Balanced event study (same cohorts pre and post)

The other 40% separately: is the effect concentrated among vulnerable mothers?

Treatment by initial stock of preschools

Sample by predicted probability of pregnancy

Appendix

Appendix A1. Full identification assumptions and placebo check

Parallel trends: assessed via pre-opening event-study coefficients; should be statistically indistinguishable from zero

No anticipation: mothers do not adjust decisions before the effective opening — depends on how much advance notice openings receive (to analyze)

Exogeneity of timing: opening timing should not correlate with unobserved UV characteristics. **Main threat**: JUNJI/Integra may prioritize UVs with higher care demand or specific socioeconomic profiles

Placebo check

An analogous event study will be estimated using the **pre-treatment birth rate** of the UV as the outcome. Non-significant pre-treatment coefficients would strengthen the parallel-trends assumption of the main model.

Appendix A2. Data limitation: UV assignment in 2014–2015

FPS (2014–2015) has no reliable UV field → those two years are anchored to the woman's **2016 UV** as a proxy

Consequence: women who left the panel before 2016 are dropped from 2014–2015 — precisely the earliest pre-trend years

RIS has not delivered a better historical crosswalk despite repeated requests; only unstandardized UV name strings are available

Idea, not yet tested: fuzzy matching on UV name strings

Appendix A3. Task status

Block	Task	Status	Note
2. Data generation	Coverage base (preschools × UV, 2014–2024)	ready	Verified; private preschools (transparency) still a possible extension
	Women panel (2014–2024, incl. FPS anchor)	ready	2014–2015 anchored to 2016 UV (see A2)
	Rentas / Cotizaciones cleaning	advanced	Independent/state/FF.AA. cotizaciones still pending
	Children registry cleaning	ready	`tuvo_hijo`, `n_hijos_menor5`
	Merge panel + outcome variables	ready	By `rut_inn`
	Treated / non-treated definition	ready	`grupo_t1`, `g1`, `muestra_t1`
3. Panel analysis	First-stage estimation (enrollment)	ready	+3pp; 2015 cohort under review
	Balance / validity check	ready	UV-level balance table
4. Data analysis	Methodology choice	ready	Sun–Abraham within Baker et al. (JEL 2026) framework
	Treatment definition	ready	Extensive margin, 0→1
	Run regressions (labor, fertility)	ready	Re-run without 2015 cohort pending
	Analyze results	in progress	Pre-trend robustness for employment
5. Mechanisms	Heterogeneity by group / level	in progress	By child age (first stage, employment)
	Public-policy implications	not started	
6. Writing	—	in progress	

Appendix A4. Balance in the fertility sample

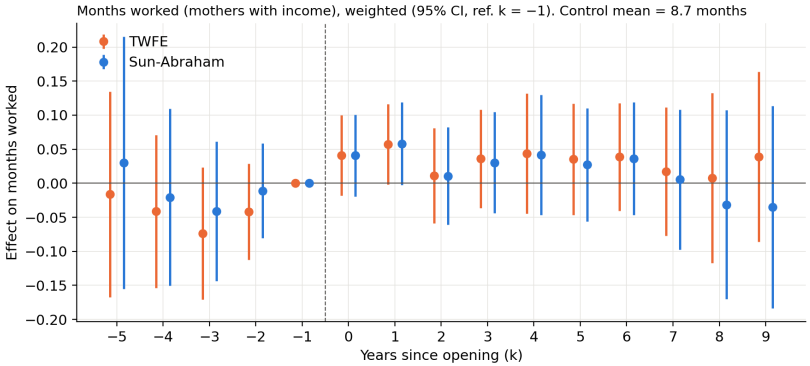
Fertility sample (women aged 18–49) — covariate levels in 2014, before any opening

	Unweighted			Weighted by UV population			
Variable	Never-tr.	Treated	Norm. diff.	Never-tr.	Treated	Norm. diff.	Sig.[a]
Birth rate	0.057	0.058	0.031	0.06	0.060	0.051	
Age	31.90	31.83	-0.053	31.74	31.82	0.095	
Children <5 per woman	0.303	0.319	0.156	0.31	0.320	0.117	**
UV population	1111.42	2025.96	**0.423**	3540.63	5305.76	**0.342**	***
Child density 0–5 UV	0.0590	0.0631	0.176	0.06	0.064	0.148	**
Women in sample per UV	233.60	445.45	**0.401**	795.29	1234.46	**0.347**	***
N (UVs)	3,973 / 422			3,973 / 422			

[a] t-test at the UV level (** $p<0.01$, *** $p<0.001$), for reference only; normalized differences prioritized (concern threshold: 0.25). UVs with at least one sample woman in 2014, hence more never-treated UVs than in the labor sample.

⇒ **Same conclusion as the labor sample: women look alike, UVs differ in size.**

Appendix A5. Months worked: no effect so far



Weighted. Mothers with earnings: about +0.03 months on a base of 8.7, clean pre-trends.

Caveat: currently conditional on working. Next: months worked with 0 for mothers without work.

Appendix A6. First stage: average effect is ≈2pp for the average child

Back to first stage

Sun–Abraham average (pp)	**Weighted (main)**	Unweighted
Post, $k = 0$–6	**1.92**	3.11 (0.58)
Post, $k = 0$–9	1.53	2.23 (0.63)
Pre, $k = -5$–-2	−0.06	0.49 (0.59)
Joint pre-test	—	$p = 0.90$
Control mean	33.2%	32.4%
Relative effect ($k = 0$–6)	**+6%**	+10%

Weighted = effect for the average child (weights: children aged 0–4 in the UV). **Unweighted** = effect for the average UV.

What changed since last meeting: the model now includes *all* event-time dummies. The negative $k = -2$ we flagged came from TWFE; it disappears with Sun–Abraham.

Average of event-study coefficients; SE (in parentheses) uses their full covariance matrix. Weighted SE pending. $k = 0$–6 is our main window (balanced across cohorts).

⇒ A clear first stage: 33% → 35%, immediate, no pre-trends — but small.

Appendix A7. Self-reported outcomes (RSH): exploratory

Outcome (mothers, RSH)	Control mean	Weighted	Unweighted
Not looking for work: no childcare	11.8%	−0.4pp	**−0.7pp***
Looking for work	18.6%	+0.6pp	+0.8pp
Weekly hours worked	36.5 h	+0.1 h	−0.3 h
Not looking for work: household chores	12.0%	not interpretable (pre-trends)	

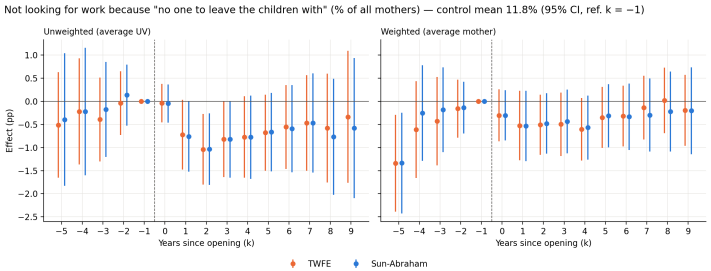
Most interesting: fewer mothers say they don't look for work because they have no one to care for their children (−6% relative), from $k = 1$, clean pre-trends — the most direct measure of the care-constraint mechanism

Weighted: same direction, weaker

Caveats: self-reported; the RSH form is not updated every year, and families may update it when applying for benefits (e.g., a preschool) → timing may depend on treatment

Sun–Abraham average effect, $k = 0$–6. * $p < 0.05$. Weighted SE pending.

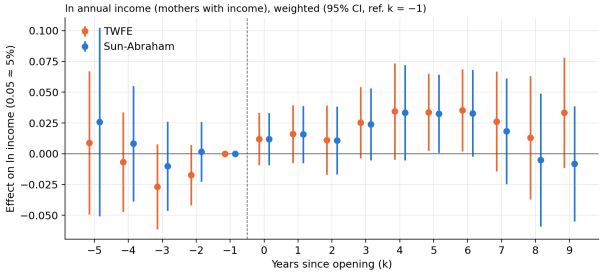
Appendix A8. RSH: fewer mothers lack childcare



Fewer mothers say they don't look for work because they have no one to leave the children with. Unweighted: -0.7pp (-6%), significant from $k = 1$, clean pre-trends; weighted: same direction, weaker

Caveats (exploratory): only mothers who neither work nor look for work answer the reason; all others are coded 0, so it is a % of all mothers — if more mothers start working, it falls too (part of the effect). The RSH form is not updated every year: the same answer can repeat for several years, so changes are slow and noisy. Some UVs have very few answers ($\sim$4 women).

Appendix A9. Previous earnings estimate (all mothers)



Same **gradual shape** as employment; clean pre-trends

$\approx$+3% at $k = 4$–6; average $k = 0$–6: **+2.3%** $\approx$ CLP 10–20k per month, not yet significant

Still to do

ln earnings is measured **only among mothers with earnings** $\rightarrow$ new entrants may pull it down

Add earnings **in levels, including zeros** for mothers without earnings

Months worked (App. A5): no effect so far, but also conditional on working $\rightarrow$ re-estimate with 0 for mothers without work

$\Rightarrow$ **Mothers who work earn $\approx$2–3% more from the fourth year: a suggestive result.**

Appendix A10. Two ways to measure earnings

Example: a UV with 4 mothers. After the opening, **C starts working**, at a somewhat lower wage (CLP million per year).

	A	B	C	D	**Earnings incl. zeros** (all mothers)	**ln earnings** (only mothers who work)
Before	4	4	0	0	$(4+4+0+0)/4 = \mathbf{2.0}$	mean of A, B $= 4$
After	4	4	**3**	0	$(4+4+3+0)/4 = \mathbf{2.75} \uparrow$	mean of A, B, C $\approx 3.6 \downarrow$

Earnings including zeros: how much income mothers have *overall* → rises if more mothers work *or* if they earn more

Expressed relative to the year's mean (100): +2 points ≈ +2% for the average mother

ln earnings: how much *working* mothers earn → can fall when new workers enter with lower wages, even if no one's wage falls

⇒ **Earnings including zeros capture new workers; ln earnings only captures wages.**